BUSINESS REQUIREMENTS DOCUMENT

Auction Management System

Technology: Django / Django REST Framework + PostgreSQL

1. Project Overview

The Auction Management System is a web-based application that allows registered users to create auction listings, browse available auction items, place bids, and participate in the auction process. An Admin/Super User will manage users, categories, auction approval, bids, and transactions.

The first version is intentionally limited to the core auction lifecycle and avoids non-critical features such as online payment gateways, complex notifications, shipping, ratings, and advanced reporting.

2. Objectives

- Provide secure registration and login for users.
- Allow users to create and manage auction listings.
- Allow users to browse auctions and place bids.
- Automatically identify the highest bidder when an auction closes.
- Maintain a basic transaction record for completed auctions.
- Provide role-based access for Admin and General User.
- Store application data using PostgreSQL.

3. User Roles

Role	Description	Main Responsibilities
Admin / Super User	System administrator with privileged access.	Manage users, categories, auctions, bids and transactions; approve/reject auctions.
General User	Registered application user.	Manage profile, create auctions, browse auctions, place bids and view own auction/bid results.

4. Authentication & User Management

4.1 User Registration

- Name
- Email
- Mobile Number
- Username
- Password
- Address

4.2 Authentication

- User Login
- User Logout
- Password Change
- Forgot/Reset Password
- Role-based authorization

4.3 User Profile

- View profile
- Edit profile information
- View personal auction and bidding activity

5. Auction Management

A registered user can create an auction item with the following information:

- Item Name
- Item Description
- Category
- Item Image
- Starting Price
- Minimum Bid Increment
- Auction Start Date/Time
- Auction End Date/Time

General User capabilities:

- Create auction
- Edit own auction
- Cancel own auction
- View own auctions

Admin capabilities:

- View all auctions
- Approve auction
- Reject auction
- Manage auction status

6. Auction Status

- Pending Approval
- Approved
- Live / Active
- Closed
- Cancelled
- Rejected

7. Auction Listing & Search

- View all available auctions
- Search auction by item name
- Filter auctions by category
- View auction details
- Display starting price, current highest bid and auction end time

8. Bidding Management

A logged-in General User can place a bid on an active auction.

The system must validate that:

- The user is authenticated.
- The auction is currently active.
- The bidder is not the seller of the auction.
- The bid is greater than the current highest bid.
- The bid satisfies the minimum bid increment.
- The auction has not expired.

8.1 Bid History

Each auction should display the previous bids, including:

- Bidder
- Bid Amount
- Bid Date/Time

9. Auction Closing & Winner

When the auction reaches its end date/time, the auction is closed and the highest valid bidder is identified as the winner.

Example:

- Item: Laptop
- Highest Bid: ₹42,000
- Winner: Highest valid bidder

After closure, the seller and winning bidder should be able to view the auction result.

10. Transaction Management

A basic transaction record will be created after an auction is completed.

- Transaction ID
- Auction
- Seller
- Buyer / Winner

- Winning Amount
- Transaction Status
- Transaction Date

Transaction Status:

- Pending
- Completed
- Cancelled

Online payment gateway integration is not included in the initial version.

11. User Dashboard

The General User dashboard should provide:

- Number of own auctions
- Number of active auctions
- Number of bids placed
- Number of winning auctions
- My Auctions
- My Bids
- Won Auctions
- Profile

12. Admin Dashboard

The Admin dashboard should provide basic summary information:

- Total Users
- Total Auctions
- Active Auctions
- Completed Auctions
- Total Bids

Admin management functions:

- Manage users
- Manage categories
- Approve/reject auctions
- View bids
- View transactions

13. Role-Based Permission

Function	Admin	General User
Register	No	Yes
Login / Logout	Yes	Yes

Edit Own Profile	Yes	Yes
Create Auction	Optional	Yes
Edit Own Auction	No	Yes
Cancel Own Auction	No	Yes
Place Bid	No	Yes
View Auctions	Yes	Yes
Approve/Reject Auction	Yes	No
Manage Users	Yes	No
Manage Categories	Yes	No
View Bids	Yes	Own/Relevant
View Transactions	Yes	Own/Relevant

14. Database Requirements

PostgreSQL will be used as the primary relational database.

Core database tables:

- Users
- Categories
- Auctions
- Bids
- Transactions

Basic relationships:

- One User can create many Auctions.
- One Auction can have many Bids.
- One User can place many Bids.
- One completed Auction can generate one Transaction.
- One User can participate in many Transactions as buyer or seller.

15. High-Level Database Structure

Users → Auctions → Bids → Transactions
Users → Bids
Categories → Auctions

16. Non-Functional Requirements

- Passwords must be securely hashed.
- Only authenticated users can place bids.
- Role-based authorization must be enforced at the backend/API level.
- Input validation must be applied to registration, auction and bidding data.
- The system should prevent invalid or duplicate business operations.
- Bid processing should use database transactions/locking where necessary to handle simultaneous bids safely.
- PostgreSQL should be used for persistent data storage.

17. Core Auction Workflow

1. User Registration
2. Login
3. Create Auction
4. Admin Approval
5. Auction Goes Live
6. Users Place Bids
7. Auction Ends
8. Highest Bidder Becomes Winner
9. Transaction Record Is Created
10. Transaction Is Completed

18. Recommended Technology Stack

Component	Technology
Backend	Python Django
API	Django REST Framework
Database	PostgreSQL
Authentication	JWT or Django Authentication
Frontend	HTML/CSS/Bootstrap or React
Database Access	Django ORM